FROM STATIC TARIFF VARIABLES TO DYNAMICALLY CONTROLLED GRID CAPACITY

An OpenADR 3.0 framework for Icelandic EV charging, distributed energy resources, storage and network flexibility

---

Master's-level technical and policy report

Prepared for stakeholder engagement with RARIK, Landsnet and related market institutions

Date: 4 August 2026

Project milestone: proof of a reliable RARIK VTN–CPO VEN interface over OpenADR 3.0 by 1 September 2026

Central proposition

The decisive transition is not the adoption of one protocol. It is the replacement of static, predefined grid variables with continuously measured, requested, allocated and verified operating variables. OpenADR 3.0 is an implementable starting interface for that transition.

Status: analytical report, not legal advice. Publicly documented facts, derived calculations, illustrative scenarios and practitioner evidence are explicitly distinguished.

Abstract

Iceland’s electricity-demand outlook, transmission investment programme and decarbonisation objectives create a timing mismatch: electricity demand and high-power charging infrastructure can develop in one to three years, while new grid infrastructure commonly requires five to fifteen years internationally and is constrained by permitting, equipment lead times, outage windows, specialised labour and finance. Landsnet projects national demand of 25.1–28.1 TWh in 2035 and 33.9–44.3 TWh in 2050, equivalent to 22–37% and 65–116% growth from current consumption, with transport electrification a major driver [1]. In parallel, Landsnet invested approximately USD 118 million in transmission development in 2025, one of its largest investment years [2].

This report investigates whether a grid architecture based on dynamically controlled variables can complement unavoidable physical reinforcement and remove avoidable barriers created by static tariff and connection assumptions. It examines Landsnet’s four-peak transmission power charge, Netmáli 1.0’s discounted-cash-flow connection-contribution method, RARIK’s current injection tariff (innmötunargjald), seasonal public charging, megawatt charging, small producers, BESS, V2G and flexible industrial demand. The report concludes that OpenADR 3.0 can serve as the interoperable VTN–VEN coordination interface for capacity requests, operating envelopes, activations and performance reports, while local protection, certified controllers and revenue-grade metering remain authoritative.

The principal finding is qualified but strong: dynamic control cannot remove energy scarcity, asset condition, protection, metering or N-1 reinforcement requirements. It can, however, convert time diversity, unused hosting capacity, storage, local production and controllable demand into an operational resource. For fully non-firm import or export, the marginal capacity cost may approach zero during unconstrained intervals, even though fixed connection and platform costs remain. The current published RARIK injection tariff does not disclose a granular cost-of-service allocation; therefore its continuing application to dispatch-only resources requires a transparent cost-causation audit rather than an assumption that every connected kilowatt creates a permanent network obligation.

The report proposes a September 2026 proof-of-interface, a dynamic operating-envelope application profile for Iceland, and a tariff reform that separates firm access, dynamic access, flexibility availability, activation and measured network benefit.

Keywords: OpenADR 3.0; dynamic operating envelope; EV charging; MCS; BESS; V2G; DSO; TSO; RARIK; Landsnet; innmötunargjald; kerfisframlag; network tariff; demand flexibility.

Executive Summary

Main conclusion

A dynamically controlled grid is not sufficient by itself, but it is a necessary component of a feasible Icelandic electrification pathway. Physical reinforcement alone cannot match the speed, geography and intermittency of emerging demand. Static allocation alone leads to rationed connections, premature capital expenditure and tariffs that annualise temporary peaks.

The bottleneck is already visible. Public charging projects, especially remote and seasonal sites, are assessed against possible maximum power and forecast utilisation even when demand is zero for long periods and the operator is willing to accept real-time limits. Megawatt Charging System (MCS) hardware can reach megawatt-scale power per port, amplifying the mismatch between installed capability and guaranteed grid demand [18].

The upstream tariff is explicitly peak-based. Landsnet's DSO capacity charge is calculated from the average of the four highest 60-minute monthly peaks at each delivery point [3]. The tariff effective 1 April 2026 is ISK 9,786,291 per annual MW [4]. A verified 1 MW reduction in all four billing peaks therefore changes the annual upstream charge by ISK 9.786 million; a 1 MW reduction in one selected peak changes it by approximately ISK 2.447 million.

Connection economics are static and asymmetric. Regulation 302/2022 requires system contributions where expected direct revenue and other benefits do not cover connection costs. The calculation is based on discounted future revenue, operating cost, capital cost and the customer contract term [7]. Netmáli 1.0 requires a power-and-energy utilisation plan, applies only 50% of revenue to the customer-specific investment in urban areas and 30% in rural areas, permits a maximum 25-year contract period, and states that the DSO calculation prevails if the parties cannot agree [8].

The exact cost basis of innmötunargjald is not publicly itemised. RARIK charges fixed annual fees plus ISK 837,638/MW based on agreed export power, while connection charges are separate [5]. RARIK historically stated that its injection tariff largely followed Landsnet and in 2022 set its power component equal to Landsnet's then tariff [6, 10]. This establishes lineage, not current cost causation. A transparent cost study is needed.

Dynamic access is a different product from firm access. A site can request preferred power and receive a time-limited upper/lower operating envelope at its connection point. The DSO controls the boundary; the CPO or site EMS chooses how chargers, BESS, V2G, HVAC and local generation comply. The site is not guaranteed its nameplate power, and the DSO is not required to micromanage equipment.

OpenADR 3.0 can be the interface. OpenADR 3.0 uses HTTP, JSON, OAuth and REST; defines programs, events, reports, VENs and resources; supports subscriptions/webhooks; and has testing/certification tooling [12]. An Icelandic application profile is still required to define capacity-request and operating-envelope semantics.

Construction remains essential, but evidence replaces speculation. Dynamic allocation does not replace energy production, protection or continuous thermal constraints. It delays or avoids reinforcement driven only by rare simultaneous peaks and provides measured evidence of where capacity is repeatedly denied. Internationally, grid projects commonly take 5–15 years while EV charging can be deployed in 1–2 years [14].

A two-sided flexibility system unlocks more than charging. Small generators, batteries and V2G resources can offer optional, dispatch-only capacity without a guaranteed export right. Large consumers can reduce net import through local variables. Compensation should distinguish availability, activation, energy, degradation/rebound and locational value.

Recommended decisions

- Complete the 1 September interface proof as a closed-loop operational test, not merely a successful API call.
- Ask Landsnet whether the four-peak DSO tariff is considered an immutable cost principle or a proxy adopted because dynamic verifiable control was previously unavailable.
- Require a cost-causation disclosure for RARIK's innmötunargjald, including the mapping of fixed and MW charges to actual cost pools and the treatment of dispatch-only resources.
- Define four separate network products: firm access, dynamic non-firm access, paid flexibility availability and paid activation.
- Use connection-point metering and local fail-safe control as the settlement and safety boundary; use OpenADR above that boundary and OCPP/EMS/device protocols below it.
- Pilot a seasonal public charging site and at least one BESS/V2G or industrial-flexibility resource to prove both import and export operating envelopes.

Contents

1. Introduction and research questions
2. Methodology and evidentiary limits
3. Icelandic tariff and institutional architecture
4. Forensic analysis of innmötunargjald
5. Demand growth, infrastructure debt and the timing gap
6. Public EV charging, seasonality and MCS
7. Static versus dynamically controlled grid variables
8. OpenADR 3.0 technical architecture
9. Flexible producers, BESS, V2G and industrial demand
10. Economic and tariff redesign
11. Quantitative scenarios
12. September proof-of-interface specification
13. Safety, cybersecurity and certification
14. Governance and implementation roadmap
15. Counterarguments and limitations
16. Conclusions and recommendations

References

Appendices

1. Introduction and Research Questions

The Icelandic electricity system is entering a period in which the central planning variable is no longer annual energy alone. Electrified transport, batteries, converter-based generation and large digital loads create power demand that is geographically concentrated, time-dependent and controllable. The traditional network model, however, converts possible maximum demand into a long-term contractual and financial obligation before the actual operating pattern is known.

The thesis examined in this report is that the principal system transition is from static predefined variables to dynamically controlled variables. OpenADR is not the thesis. It is a practical interface through which the thesis can first be demonstrated between a DSO virtual top node (VTN) and a customer or aggregator virtual end node (VEN).

Research proposition

Where a customer does not request a permanent firm entitlement and grants the DSO enforceable control of a time-varying connection-point envelope, the network service, cost causation and tariff treatment are materially different from a conventional firm connection.

1.1 Research questions

1. What costs can be verified as being recovered by RARIK's current innmötunargjald, and which cost allocations remain undisclosed?
2. How do Landsnet's four-peak power tariff and Netmáli 1.0's discounted-cash-flow model propagate static assumptions into DSO tariffs and connection contributions?
3. Can OpenADR 3.0 provide a reliable two-way interface for capacity requests, operating envelopes, activation and performance verification?
4. How would dynamic allocation change the treatment of seasonal EV charging, MCS, small generators, BESS, V2G and large flexible consumers?
5. Which physical, safety and economic costs remain even under fully non-firm operation?
6. What can be proven by the proposed RARIK VTN milestone on 1 September 2026, and what cannot?

1.2 Scope

The report focuses on the operational and tariff boundary among Landsnet, RARIK and a CPO/aggregator. It treats ELMA or other market platforms as energy-trading and settlement interfaces rather than replacements for grid control. It does not provide a full power-flow study, legal opinion, cyber-penetration test or investment-grade business case. Numerical scenarios labelled "illustrative" are constructed to expose mechanisms, not to predict a specific site.

2. Methodology and Evidentiary Limits

The analysis combines: (i) current Icelandic tariffs and regulations; (ii) official demand forecasts and financial reporting; (iii) OpenADR Alliance documentation; (iv) international evidence on dynamic operating envelopes and flexible connections; (v) official tourism data as a national seasonality proxy; and (vi) practitioner evidence supplied in the underlying project discussion.

Evidence class	Treatment in this report
Publicly documented fact	Cited to an official tariff, law, regulator, system operator, standards body or government dataset.
Derived calculation	Arithmetic consequence of a cited tariff or dataset; assumptions are shown.
Illustrative scenario	Synthetic values used to test mechanisms; never presented as an observed site.
Practitioner evidence	Operational experience reported by the project proponent; identified as such when not found in public documents.
Inference	Reasoned conclusion from multiple sources; explicitly described as an inference rather than a disclosed fact.

2.1 Important evidentiary qualification: 630 A and lease-term practice

The practitioner account states that connection applications above approximately 630 A are commonly subjected to a DSO return-on-investment assessment using a binding site lease and a representative 24-hour utilisation forecast. The public regulation and Netmáli establish the discounted-revenue method and the use of contract term

and time-series utilisation, but the authors did not identify a statutory 630 A threshold or an express legal requirement that the contract term must be evidenced by a property lease [7, 8]. The threshold and lease implementation should therefore be documented through RARIK's application procedures, correspondence or calculation workbooks before being presented as a legal rule.

3. Icelandic Tariff and Institutional Architecture

3.1 The formal cost pipeline

The electricity value chain can be represented as generation and wholesale energy, transmission and system operation, distribution, retail/CPO activity and end use. The tariff pipeline is not identical to the physical power flow, but upstream transmission charges are incorporated into DSO revenue requirements and passed into downstream tariffs. Icelandic regulatory material confirms that transmission cost paid by DSOs is added to DSO revenue limits as actual cost [21].

The project proponent's institutional thesis is that Landsvirkjun occupies the practical top of the "food chain" because of its generation position and influence, even though formal ownership, statutory duties and tariff regulation separate Landsvirkjun, Landsnet and the DSOs. This report treats that claim as a stakeholder-power hypothesis to test in interviews, not as a conclusion proven by public corporate diagrams.

3.2 Landsnet's static power variable

Landsnet charges each DSO delivery point an annual delivery charge, a power charge and an energy charge. The power charge is based on the average of the four highest 60-minute monthly power peaks in the calendar year [3]. As of 1 April 2026, the DSO power tariff is ISK 9,786,291 per annual MW, excluding VAT [4].

$$P_{bill} = (P_{peak,1} + P_{peak,2} + P_{peak,3} + P_{peak,4}) / 4$$

The static billing variable is determined after selecting four monthly one-hour peaks. It does not directly distinguish controllable from uncontrollable load during those hours.

Current Landsnet DSO tariff component	Value effective 1 Apr 2026
Delivery charge	ISK 9,531,530 per delivery point per year
Capacity charge	ISK 9,786,291 per annual MW
Energy charge	ISK 708.64 per MWh
System services	ISK 122.04 per MWh
Transmission losses	ISK 65.52 per MWh

Landsnet's tariff already recognises a separate category of curtailable transmission and states that no capacity charge is collected when its B5 conditions are satisfied [4]. This is important precedent: the tariff system already accepts that interruptible access is economically distinct from firm capacity, even though the proposed dynamic model is more granular and interactive.

3.3 Netmáli 1.0 and connection contributions

Regulation 302/2022 requires a DSO to calculate and collect a system contribution when expected direct revenue and other benefit do not cover the cost of the connection. The formula is based on discounted future revenue, future operating cost, capital cost and connection charge; the discount horizon for users is the contract term [7]. Netmáli 1.0, effective for all DSOs from 15 May 2025, provides the common calculation method [8, 9].

Netmáli design variable	Published rule
Utilisation plan	Power and energy plan over ramp-up and expected contract term, with time series aligned to Landsnet B7.
Maximum contract term	25 years.
Revenue share credited in urban area	50% to customer-specific connection investment; 50% allocated to backbone system.
Revenue share credited in rural area	30% to customer-specific connection investment; 70% allocated to backbone system.
Disagreement on utilisation	The DSO calculation applies if the parties do not agree.
Additional cost	Added to connection charge and expected usage charges;

Netmáli design variable	Published rule
	calculated using discounted cash flow.

Additional contribution = −NPV(investment) + NPV(revenue share × tariff revenue − operating cost)

A negative result represents the estimated shortfall charged to the applicant. Under a static model, controllability has no explicit value unless the DSO incorporates it into the investment and revenue assumptions.

3.4 Why static variables cascade

The upstream four-peak charge incentivises the DSO to manage annual coincident peaks. The connection model then asks an applicant to forecast utilisation over a long contract term so the DSO can estimate whether tariffs recover investment. Where a remote charging site has low current use but strategic seasonal value, the same characteristic is interpreted twice as a liability: high possible peak power increases the technical investment, while low average energy reduces expected tariff revenue.

4. Forensic Analysis of Innmötunargjald

4.1 What is publicly known

RARIK's current injection tariff, version 46 effective 1 December 2025, applies to generating units above 16 A that inject into the distribution network. It contains fixed annual fees by generator class and an annual capacity fee of ISK 837,638/MW excluding VAT. Both fixed and capacity charges are based on the agreed export power in the connection and parallel-operation agreement. The physical connection charge is billed separately [5].

Generator category	Annual fixed fee excl. VAT
Asynchronous, below 35 kW	ISK 105,930
Asynchronous, 35–59 kW	ISK 141,240
Asynchronous, 60–99 kW	ISK 200,090
Synchronous generator	ISK 294,250
Capacity component	ISK 837,638 per MW per year

RARIK also calculates payments to qualifying smaller generators for the benefit of reduced withdrawal from Landsnet. The power component uses the measured 60-minute peak and the difference between the average of the four highest monthly peaks with and without the generator [5]. That mechanism explicitly acknowledges that distributed generation can create an upstream network benefit.

4.2 Numerical burden on small resources

For an asynchronous 20 kW producer, the published annual injection fee is ISK 105,930 + 0.02 × ISK 837,638 = ISK 122,683 excluding VAT. One hundred identical resources represent 2 MW in aggregate and would pay approximately ISK 12.27 million annually before connection charges. This calculation does not prove that the tariff is excessive; it shows how a fixed site charge can dominate the economics of small, intermittently used resources.

4.3 What is not publicly disclosed

Direct answer to the cost question

The public RARIK tariff does not itemise how much of innmötunargjald is assigned to metering, protection, control-room operation, network capacity, losses, administration, depreciation, return on capital or other regulated cost pools. The actual granular cost allocation therefore cannot be determined from the published tariff alone.

General DSO revenue limits are built from permitted capital cost, operating cost and depreciation; transmission cost and network losses are also passed into DSO revenue requirements [20, 21]. Those categories establish what the monopoly network as a whole may recover. They do not establish which marginal cost is caused by a dispatch-only 20 kW battery or V2G resource at a particular feeder and time.

4.4 Historical lineage and the “ancient tariff” hypothesis

RARIK stated in 2009 that its injection tariff largely followed Landsnet’s tariff [10]. In 2022 it stated that the revised injection tariff was aligned with Landsnet and that the capacity charge would be the same as Landsnet’s [6]. This supports the hypothesis that the tariff evolved from an upstream static allocation model rather than from a contemporary dynamic cost-to-serve study.

However, historical lineage is not proof that every component is irrelevant. The defensible academic conclusion is narrower: dynamic dispatch changes the cost-causation question, while the published tariff does not disclose enough information to test whether the current MW charge reflects the remaining cost. A cost-causation audit is required.

4.5 Supreme Court evidence: tariff allocation versus total cost

In 2024 the Supreme Court confirmed that Landsnet was not authorised to levy its transmission in-feed capacity fee on producers. Landsnet described the charge as a reallocation among customer groups rather than an increase in total revenue [11]. Landsnet’s 2025 financial statements report that approximately USD 22.4 million was refunded and collection shifted to end users—large users and DSOs—because the total revenue cap was unchanged [2].

This does not invalidate RARIK’s distribution-level injection tariff. It does demonstrate that an “in-feed charge” can operate as a revenue-allocation mechanism rather than a direct invoice for incremental physical cost. It strengthens the demand for transparency: what exact service is the distribution injection tariff purchasing, and how does that service change when export exists only on DSO request?

4.6 Cost-causation audit required from RARIK

- Identify every ledger cost pool recovered through the fixed and MW components.
- Separate one-time connection and protection costs from recurring annual costs.
- Show whether the MW component represents reserved firm export capacity, shared assets, losses, administration or an allocation of upstream charges.
- Quantify avoided Landsnet capacity and energy charges by location and hour.
- Demonstrate the incremental cost of a resource with zero firm export entitlement and a hardware-enforced zero-export fallback.
- Disclose whether the tariff is reconciled against actual utilisation, curtailment and network benefit.
- Explain why a capacity charge should apply to export that the DSO can refuse before it occurs.

5. Demand Growth, Infrastructure Debt and the Timing Gap

Landsnet forecasts national electricity demand of 25.1–28.1 TWh in 2035 and 33.9–44.3 TWh in 2050, representing 22–37% and 65–116% growth respectively. Transport electrification is identified as a principal driver [1]. The forecast also identifies persistent near-term energy shortage through 2029 under its assumptions.

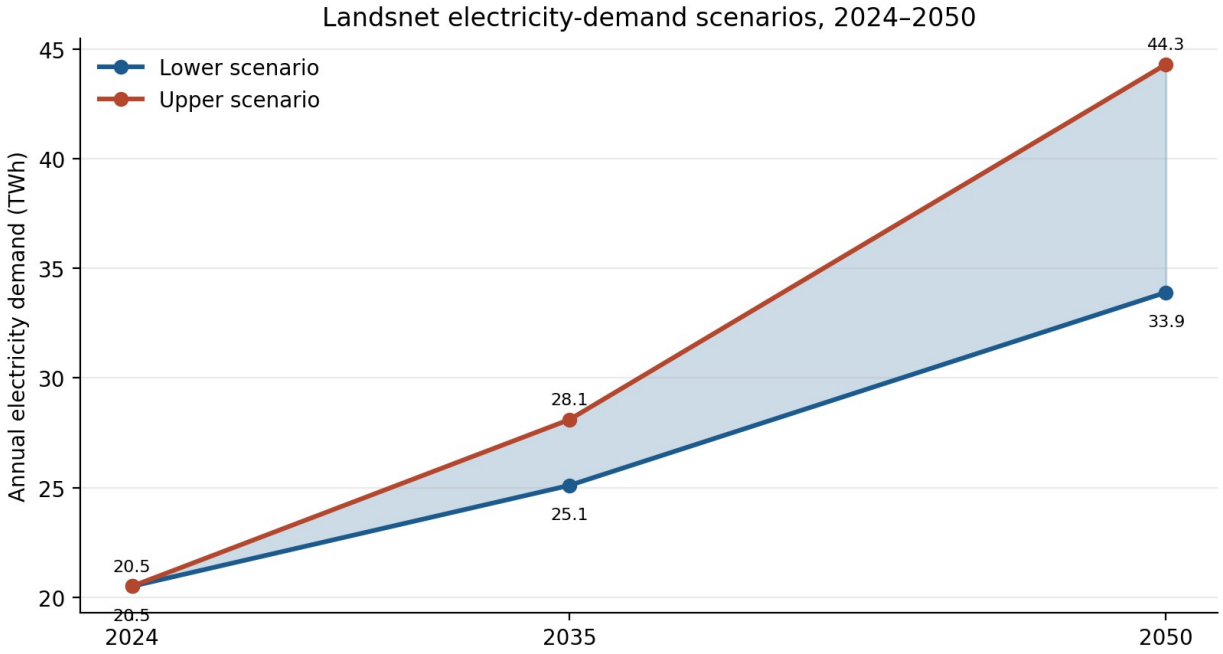


Figure 1. Landsnet demand range and implied current baseline.

Source: Landsnet 2024–2050 electricity outlook [1]. The 2024 baseline is approximately inferred from the published growth percentages.

Landsnet's 2025 transmission investment was approximately USD 118 million, one of the company's largest investment years [2]. High investment therefore does not contradict the bottleneck thesis; it confirms the scale of physical work already under way. The bottleneck is a rate problem: demand, charging equipment and storage can appear faster than lines, substations and permitting processes can be completed.

The IEA reports that planning, permitting and completing new grid infrastructure can take 5–15 years, compared with 1–2 years for EV charging infrastructure [14]. Icelandic project lead times vary, but the structural mismatch is directly relevant: even an unconstrained investment budget cannot instantly create qualified engineering hours, construction windows, transformers, switchgear, cable supply and statutory approvals.

5.1 Three distinct scarcity problems

Scarcity	Can dynamic control solve it?	Required response
Annual energy (MWh)	Only partly. It can shift and reduce demand but cannot create sustained energy.	New generation, efficiency, fuel substitution and market balancing.
Coincident capacity (MW)	Often substantially. Timing, storage, local generation and load control can release headroom.	Dynamic envelopes, flexibility procurement, storage and targeted reinforcement.
Reliability / redundancy	Not where an independent physical route or N-1 capability is required.	Lines, substations, protection, condition renewal and emergency resources.

The policy error would be to present software as a replacement for steel. The equally serious error is to assume steel alone can be delivered fast enough and therefore price every possible peak as firm capacity. Dynamic control is the bridge between current assets and the long-run reinforced network.

6. Public EV Charging, Seasonality and MCS

6.1 The seasonal-site paradox

The locations with the weakest grid can also be the locations with the strongest seasonal transport need. A remote public charging site may require almost no power during long periods but become nationally important during summer travel, holidays or disruption on alternative routes. A static power tariff annualises a small number of high-demand hours; a utilisation-based ROI model then penalises the same site for low average throughput.

Official Icelandic Tourist Board data provide a national—not site-specific—seasonality proxy. In 2025, foreign departures through Keflavík ranged from 120,016 in December to 312,964 in August, a 2.61× ratio. The June–August average was 2.18× the December–February average [19]. This does not measure charging demand directly, but it substantiates the premise that important Icelandic transport demand is highly seasonal.

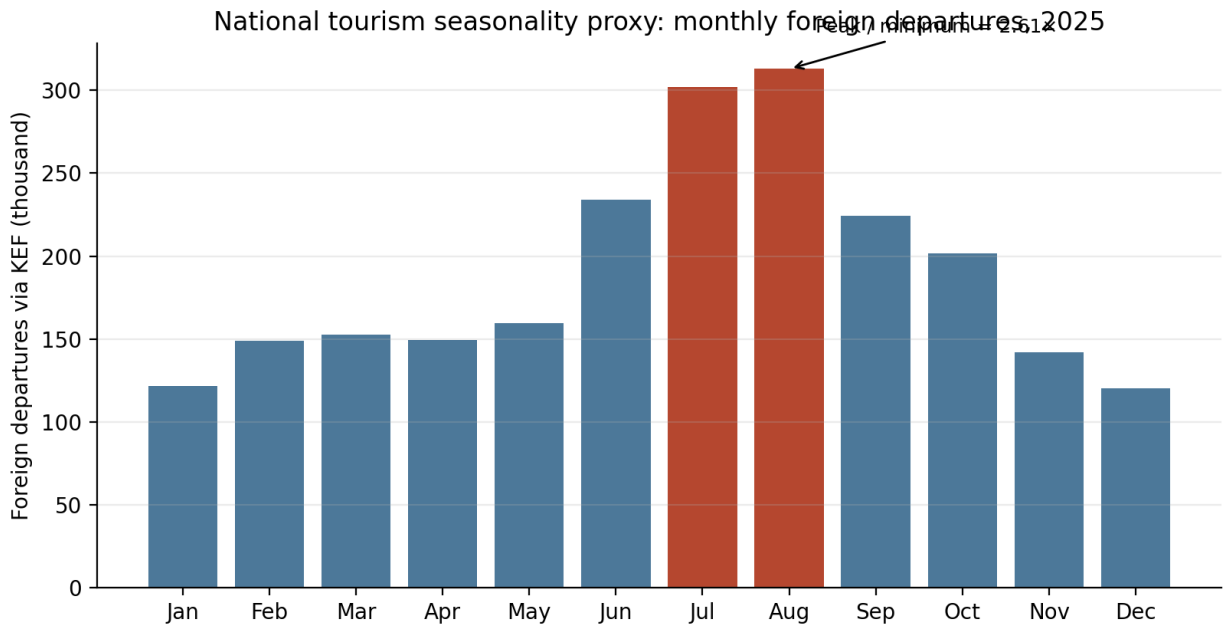


Figure 2. Tourism seasonality proxy relevant to remote public charging.

Source: Icelandic Tourist Board workbook, monthly foreign departures via Keflavík, 2025 [19].

6.2 The self-reinforcing market failure

The static model can create a circular failure: no existing EV traffic produces a weak utilisation forecast; the weak forecast produces a high connection contribution; the contribution prevents a bankable charging site; the absence of charging prevents EV adoption and traffic; the initial low forecast then appears validated. This is not a neutral forecast error. It can lock in the infrastructure conditions that the forecast assumed.

Perverse incentive

The current method can reward installation of yesterday's charging power because future-proof equipment increases theoretical peak exposure before demand exists. Installed charger capability, guaranteed grid capacity and actual time-dependent demand are treated as though they were the same variable.

6.3 MCS magnifies the mismatch

The IEA describes MCS maximum power using specifications of 1,250 V and 3,000 A, corresponding to multi-megawatt theoretical capability [18]. A truck-charging hub can therefore have several megawatts of installed port capability while actual simultaneous demand remains intermittent and develops over time. Treating nameplate sum as a permanent grid obligation can make public heavy-transport charging non-bankable before the fleet has an opportunity to transition.

A dynamic connection separates equipment readiness from network entitlement. A hub can install high-power chargers, receive a low or zero firm allocation, request additional capacity when a truck connects, and combine the DSO allocation with local BESS. Repeated unserved requests then become objective evidence for reinforcement.

7. Static versus Dynamically Controlled Grid Variables

7.1 The static system

Variable	Static interpretation	Consequence
Connection power	Long-term maximum or nameplate proxy	Capacity reserved or reinforced before use is observed.
24-hour utilisation profile	Representative truth over contract period	Forecast error becomes an applicant financing risk.
Producer export capacity	Annual agreed MW obligation	Dispatch-only resources are charged as if export were continuously available.
Site identity	Consumer or producer	BESS/V2G bidirectionality is poorly represented.
Network value	Average tariff category	Time and location of relief are weakly valued.
Investment signal	Application forecast	Reinforcement may precede persistent measured scarcity.

7.2 The dynamic system

The dynamic model defines the connection point as a controlled operating range. The site reports what it wants and what it can flex; the DSO returns what the network can safely support for a defined validity period. The DSO rents available network capability rather than guaranteeing the installed equipment rating.

$$P_{export,min}(t) \leq P_{connection}(t) \leq P_{import,max}(t)$$

The connection-point power must remain within time-varying lower and upper bounds. Sign convention can be defined in the Icelandic profile; the principle is a bidirectional dynamic operating envelope.

$$P_{allocated}(t) = \min[P_{requested}(t), H_{network}(t)]$$

For a simple import request, allocated power is limited by the network hosting capacity H at the relevant location and time. More advanced allocation includes fairness, priority, contingencies and reactive power.

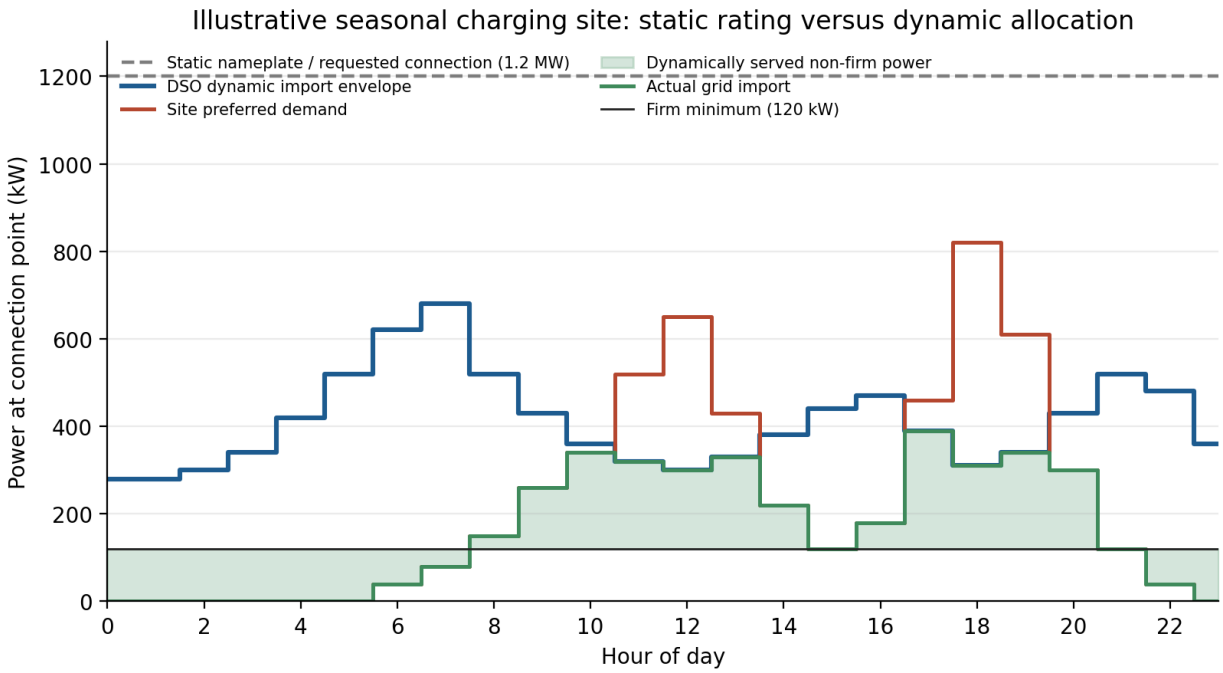


Figure 3. Illustrative static rating and dynamic hourly allocation.

Source: Synthetic example prepared for this report. It is not measured site data.

7.3 Request–response cycle

Site / CPO report	DSO / VTN response
Current net import/export	Maximum permitted import and export
Preferred power and minimum useful allocation	Validity interval and ramp requirement
Required energy and deadline/departure	Priority and curtailment status
Available up/down flexibility	Activation price or contract reference
BESS state of charge and constraints	Fallback envelope and communication timeout
Measured performance	Acceptance, exception or revised envelope

7.4 The DSO controls the boundary, not the customer's technology

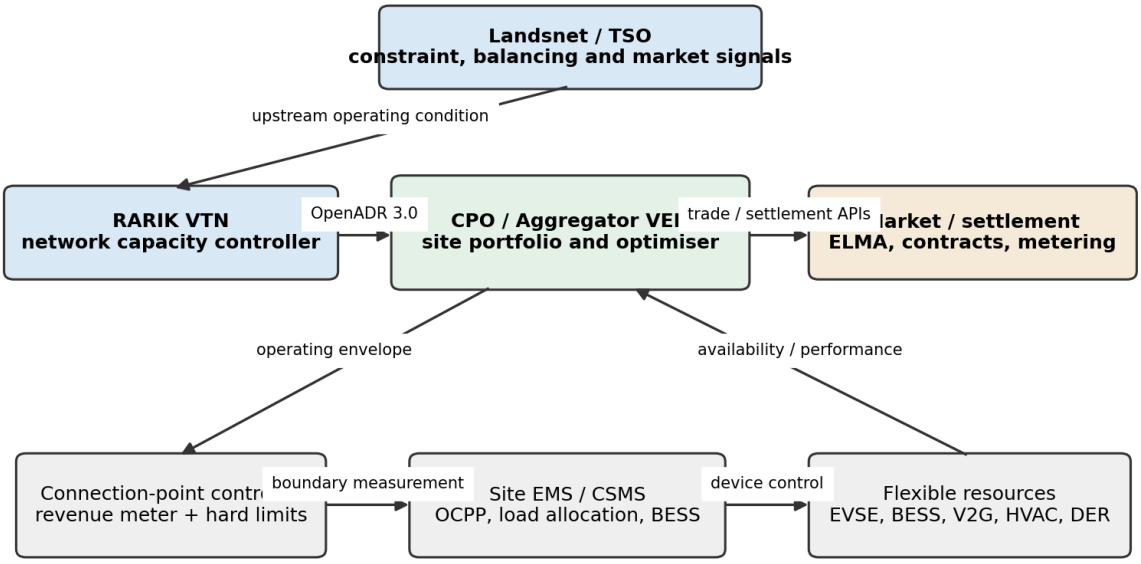
The DSO should not select whether the customer uses a battery, charger queue, HVAC adjustment, onsite generator or process change. It should define and enforce the connection-point operating envelope. The customer or aggregator remains responsible for internal optimisation, subject to certified equipment and contractual constraints.

This distinction resolves the earlier "technology-neutral" debate. The DSO is not purchasing a behind-the-meter technology solution; it is controlling use of shared network equipment and, when required, purchasing a verified boundary outcome.

8. OpenADR 3.0 Technical Architecture

8.1 Suitability as the dynamic interface

OpenADR 3.0 uses HTTP, JSON, OAuth and a RESTful architecture. It introduces program management, APIs for VENs and resources, events and reports, subscriptions/webhooks, model extensions and testing/certification tools [12]. These elements are sufficient to implement the proposed capacity dialogue without redefining the core transport protocol.



Safety and protection remain local; OpenADR coordinates capacity, events and reporting.

Figure 4. Proposed protocol and responsibility architecture.

Source: Author's synthesis based on OpenADR 3.0 objects and the Icelandic stakeholder model.

8.2 Role mapping

OpenADR concept	Proposed Icelandic use
VTN	RARIK dynamic-capacity server; potentially Landsnet at a separate upstream layer.
VEN	CPO, aggregator or site controller representing one or more connection-point resources.
Program	Commercial and operational rules: dynamic access, congestion relief, TSO activation, emergency event.
Resource	Connection point, site, portfolio or certified sub-resource with grid-location metadata.
Report	Current power, requested power, forecast, flexibility availability, state of charge, measured response.
Event	Import/export envelope, activation request, price signal, priority and validity interval.
Subscription	Near-real-time delivery of changed events and reports through webhook/push mechanisms.

8.3 Required Icelandic application profile

OpenADR does not define a universal message named "request X kW and return available Z kW." The project should define a constrained application profile so both parties interpret the same attributes and units. OpenADR 3.0 supports agreed extensions and private strings while preserving core behaviour [12].

Profile object / signal	Minimum required fields
IS_IMPORT_CAPACITY_REQUEST	Resource ID; current kW; preferred kW; minimum useful kW; required kWh; start/deadline; confidence.
IS_OPERATING_ENVELOPE	P_import,max; P_export,max; start; end; ramp; priority; fallback; reason code.
IS_FLEXIBILITY_OFFER	Up/down kW; duration; energy limit; response time; price; availability confidence.
IS_ACTIVATION	Requested delta-kW or boundary target; start; duration; tolerance; compensation reference.
IS_NET_POWER_REPORT	Revenue-meter kW/kWh; timestamp; quality flag; event/program IDs.

Profile object / signal	Minimum required fields
IS_EXCEPTION	Unavailable resource; local safety override; communications degradation; reason and expected recovery.

8.4 Protocol boundaries

Layer	Primary interface / responsibility
Substation protection and grid control	SCADA, protection relays, IEC 61850 where applicable; not OpenADR.
Connection-point enforcement	Certified local controller, meter and hard import/export limits.
DSO–CPO / aggregator coordination	OpenADR 3.0 VTN–VEN.
Charger management	OCPP / CSMS.
Vehicle–charger communication	ISO 15118 and MCS/charging standards.
Inverter/BESS local control	Vendor EMS, Modbus/industrial protocols and applicable grid-code functions.
Energy market and financial settlement	ELMA/market APIs, contracts and settlement systems.

OpenADR should therefore be described as the digital nervous system, not the protection reflex. A remote event can never override local protection, anti-islanding or certified equipment limits.

9. Flexible Producers, BESS, V2G and Industrial Demand

9.1 Dispatch-only resources

A small producer, BESS owner or V2G vehicle owner need not receive any guaranteed export right. The resource can declare availability and export only when the DSO or TSO calls it under an agreement. The network is purchasing an option and an activation outcome, not reserving continuous export capacity for the owner.

Payment component	Economic purpose
Availability	Compensates the owner for keeping a defined capability ready and not selling it elsewhere.
Activation	Pays for the actual change in boundary power when called.
Energy	Pays for exported or additionally consumed kWh.
Asset impact	Battery degradation, efficiency losses, comfort/process effects and rebound.
Locational value	Recognises relief at a constrained feeder, transformer or Landsnet delivery point.

9.2 V2G relevance

V2G increases the number and geographic distribution of potential resources, but it also makes static classification less defensible. The same connection can consume, reduce consumption or export depending on vehicle state of charge, driver departure requirements and network need. The resource should be characterised by an available operating range and energy budget, not a permanent producer label.

A V2G report can include maximum export kW, available energy kWh, minimum state of charge, departure time and response time. The DSO activates only the amount useful at that grid location; the aggregator selects vehicles and preserves driver constraints.

9.3 Large consumers as boundary flexibility

A large consumer can use its own production and storage for normal operations and offer net-load change when grid demand arises. The DSO requests an outcome such as “reduce net import by 400 kW for 30 minutes.” The site can combine reduced charging, HVAC adjustment, BESS discharge and local generation while maintaining internal constraints.

$$F_{delivered}(t) = P_{baseline}(t) - P_{actual}(t)$$

For import reduction, delivered flexibility is the difference between an agreed baseline and measured actual import. Baseline governance and rebound treatment are essential for settlement.

9.4 DSO–TSO coordination and double commitment

A resource can be locally valuable to RARIK and systemically valuable to Landsnet, but the same kW cannot be sold twice for overlapping obligations. A hierarchy or “primacy” rule is needed. Mandatory local safety envelopes should dominate commercial TSO activation; available residual flexibility can then be offered upstream. Every event and commitment should carry resource, location, time, priority and exclusivity identifiers.

10. Economic and Tariff Redesign

10.1 Four separate products

Network product	Customer right / obligation	Tariff logic
Firm access	Guaranteed capacity except defined emergencies.	Full capacity-related charge and attributable reinforcement.
Dynamic non-firm access	Capacity varies within an operating envelope; curtailment accepted.	Low fixed platform/service fee plus measured congestion or access charge.
Flexibility availability	Resource remains ready under agreed limits.	Availability payment.
Flexibility activation	Resource changes net power when called.	Activation, energy and asset-impact compensation.

10.2 Proposed dynamic tariff identity

Network charge = fixed service + firm capacity + measured congestion + ICT/settlement − verified network benefit

The formula is conceptual. Cost pools must be calibrated through a regulated cost-of-service study and must preserve revenue adequacy without charging non-firm resources as firm capacity.

The dynamic model does not promise a zero bill. It changes which costs are attributable. Physical connection, protection, metering, communications, cyber operations, administration and shared network service remain. The avoidable component is the permanent reservation and premature reinforcement associated with capacity the customer neither requests nor receives as a firm right.

10.3 When marginal capacity cost can approach zero

For a fully controllable resource, the marginal capacity cost can approach zero when all of the following hold: no firm import/export right is granted; the resource is admitted only inside measured headroom; a local controller enforces the limit; communication failure produces a safe fallback; no reinforcement is undertaken for the non-firm portion; and the DSO can refuse or curtail the flow before a constraint occurs.

The total service cost is still positive. “Close to zero” is therefore an economically precise claim only for incremental network-capacity cost in unconstrained intervals, not for the entire connection or system.

10.4 Reforming the connection ROI calculation

The investment counterfactual should be the least-cost service actually requested, not the full technical rating of installed charging equipment. A dynamic assessment compares the investment required for the guaranteed firm envelope with the cost and value of a non-firm layer, storage and operational control.

Dynamic contribution = unavoidable incremental investment − NPV(tariff income + flexibility value + deferred-investment value + avoided upstream cost)

The equation expands the current discounted-cash-flow method to include measurable benefits and investment deferral. It does not require the DSO to forecast customer technology; it requires the DSO to value the boundary service it controls.

10.5 Symmetry and reconciliation

The applicant currently bears downside risk when utilisation is forecast to be low. A fair dynamic model should include periodic reconciliation when actual tariff revenue, denied-capacity hours or later shared use materially

depart from the original assumptions. Regulation 302/2022 already provides for cost sharing or repayment when additional users use a reinforced element within ten years [7]. The same principle can be extended to actual utilisation and flexibility value.

11. Quantitative Scenarios

11.1 Upstream peak value

Verified change at a Landsnet DSO delivery point	Arithmetic change in annual capacity charge
1 MW reduction in each of the four selected billing peaks	1.00 × ISK 9,786,291 = ISK 9.786 million
1 MW reduction in only one selected peak	0.25 × ISK 9,786,291 = ISK 2.447 million
0.5 MW reduction in each of four selected peaks	0.50 × ISK 9,786,291 = ISK 4.893 million

These values are direct tariff arithmetic, not a prediction of savings passed to a specific site. Whether and how a DSO shares the avoided cost depends on tariff reform, delivery-point coincidence and the resource's verified performance.

11.2 Illustrative remote charging hub

Parameter	Static model	Dynamic model
Installed EVSE capability	1.2 MW	1.2 MW
Guaranteed grid right	1.2 MW assumed in planning	120 kW firm; 0–1.08 MW non-firm
Demand when no vehicle is present	Capacity remains financially relevant	Request and allocation may be near zero
Peak service	Full grid import if available	DSO envelope + optional 500 kW / 1 MWh BESS
Investment signal	Forecast peak and contract utilisation	Measured denied-request hours and persistent congestion
Failure mode	Breaker/static contract limit	Local fail-safe fallback to 120 kW or zero non-firm component

In the synthetic profile in Figure 3, the 1.2 MW equipment rating is never reached. The DSO envelope serves part of the preferred request, while the customer can decide whether to extend dwell time, use local storage or communicate an expected service level. The system obtains real demand evidence without first guaranteeing a permanent 1.2 MW entitlement.

11.3 Small-resource portfolio

A portfolio of one hundred 20 kW asynchronous producers or batteries has 2 MW aggregate nameplate capability. Under the current RARIK fee, the annual fixed-plus-capacity amount is approximately ISK 12.27 million excluding VAT before connection charges. If the portfolio is dispatch-only and exports only when authorised, its economic service resembles an availability option. The key tariff question becomes which recurring cost remains after firm export reservation is removed.

11.4 What the scenarios do and do not prove

The scenarios prove arithmetic and demonstrate design logic. They do not prove feeder headroom, customer willingness, battery economics, market prices or the exact amount of deferred reinforcement. Those require site power-flow data, interval metering, network topology, outage criteria and a regulated benefit-sharing rule.

12. September Proof-of-Interface Specification

The proposed milestone is a reliable OpenADR 3.0 interface with RARIK operating a VTN. The proof should be defined as a closed-loop boundary-control test. A demonstration that merely creates an OpenADR event proves syntax and authentication, not dynamic grid capacity.

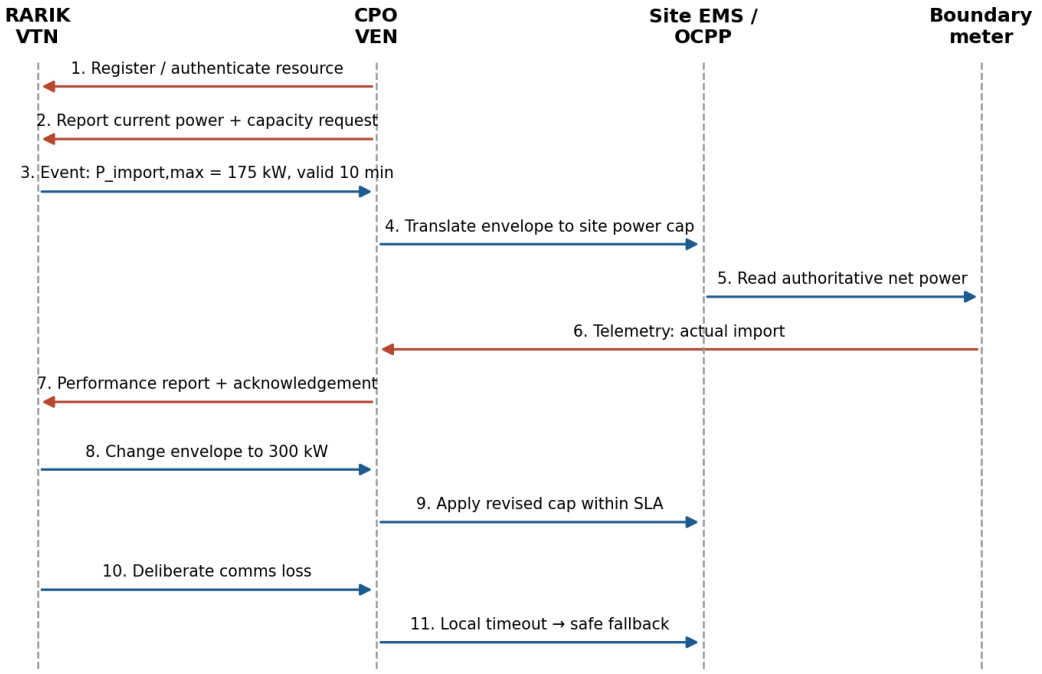


Figure 5. Proposed 1 September closed-loop proof sequence.

Source: Author's test specification.

12.1 Minimum test cases

Test	Procedure	Pass criterion
Authentication and resource registration	VEN authenticates to VTN and resource IDs are reconciled.	No unauthorised access; identical resource mapping on both sides.
Capacity request	VEN reports current, preferred and minimum useful kW.	VTN receives valid, time-stamped report.
Operating envelope	VTN sends 175 kW maximum for a ten-minute interval.	VEN acknowledges and local site cap is applied.
Physical response	Simulated or real load attempts to exceed 175 kW.	Boundary meter remains within tolerance.
Envelope revision	VTN changes limit to 300 kW.	Site applies change within agreed response SLA.
Telemetry and audit	Metered kW returned with event/program IDs.	RARIK and CPO logs reconcile within clock tolerance.
Communications loss	Network path deliberately interrupted.	Local controller enters agreed safe fallback before timeout.
Recovery	Communication restored and new event issued.	No uncontrolled jump; explicit resynchronisation and acknowledgement.

12.2 Proposed quantitative KPIs

KPI	Initial proof target
Interface availability during test window	≥99.5% excluding planned interruption
Clock synchronisation error	≤1 second
Event delivery and acknowledgement	≤5 seconds on normal network
Site cap application	≤10 seconds after valid event
Boundary power compliance	Within ±5% or agreed meter/control tolerance
Telemetry reporting interval	5–10 seconds during active event
Communication-loss fallback	Within 30–60 seconds, contract-defined
Audit completeness	100% of requests, events, acknowledgements, commands and

KPI	Initial proof target
	meter values linked by IDs

12.3 What to exclude from the first proof

- Full market pricing and financial settlement.
- TSO–DSO primacy across multiple simultaneous programmes.
- Fleet-scale optimisation across many sites.
- V2G export unless already certified and safe.
- Claims of avoided capex without a network study.

Keeping these outside the critical path does not reduce the importance of the proof. It isolates the fundamental proposition: a DSO can dynamically allocate connection-point power without controlling individual chargers and can verify compliance through the boundary meter.

13. Safety, Cybersecurity and Certification

13.1 Non-negotiable local authority

The connection-point controller and protection system remain authoritative. OpenADR events are supervisory and commercial instructions. They may narrow an operating range but may not defeat local overcurrent, voltage, frequency, anti-islanding, thermal, earthing or emergency-stop functions.

13.2 Fail-safe state

Every dynamic connection requires a contractually defined communication timeout and fallback. Options include zero non-firm import/export, a conservative static firm limit, or the last valid envelope for a short grace period. The fallback must be enforced locally even if the VEN, cloud platform or telecommunications network fails.

13.3 Security controls

OpenADR 3.0 uses TLS and OAuth 2.0 credential flow; the Alliance provides VTN/VEN testing and an audit trail [12]. Production deployment additionally requires credential lifecycle management, least privilege, network segmentation, event signing or integrity controls where appropriate, time synchronisation, immutable logs, rate limits, incident response and separation between commercial APIs and protection networks.

13.4 Certification perimeter

Certification / assurance domain	Purpose
OpenADR VTN/VEN interoperability	Confirms conformance of protocol implementation and object exchange.
Connection controller and metering	Confirms accurate boundary enforcement and settlement-grade measurement.
EVSE and vehicle interfaces	Confirms charger safety, OCPP behaviour and applicable ISO 15118/MCS functions.
BESS/inverter grid compliance	Confirms export limits, anti-islanding and grid-support behaviour.
Site commissioning test	Confirms the integrated system follows envelopes and enters safe fallback.
Operational audit	Confirms ongoing availability, performance and cyber controls.

14. Governance and Implementation Roadmap

14.1 Stakeholder roles

Stakeholder	Proposed role
Landsnet	Define upstream constraint/value interface; recognise verified DSO peak flexibility; coordinate system services and primacy.
RARIK	Operate VTN; calculate local operating envelopes; procure local flexibility; disclose tariff cost causation.
CPO / aggregator	Operate VEN; aggregate sites/resources; preserve customer requirements; control devices through EMS/OCPP.
Raforkueftirlit / regulator	Approve tariff/access products, cost allocation, consumer

Stakeholder	Proposed role
	protections and revenue-recovery transition.
Equipment owners	Opt in under clear availability, activation and asset-impact terms.
Market/settlement operator	Handle energy trades, imbalance and financial settlement outside the safety/control path.

14.2 Phased roadmap

Phase	Deliverable
Phase 0 – 1 September proof	One VTN, one VEN, one resource, capacity request, changing envelope, meter report and deliberate communications failure.
Phase 1 – controlled field pilot	Seasonal public EV site; real boundary meter; operational but non-financial envelopes; weekly performance review.
Phase 2 – two-sided flexibility	Add BESS or small generator; test optional export and paid activation; establish baselines and degradation compensation.
Phase 3 – Landsnet interface	Aggregate resources by delivery point; test an upstream peak-risk window and RARIK allocation to local resources.
Phase 4 – tariff sandbox	Firm/non-firm tariff, avoided upstream cost credit, connection-contribution reconciliation and independent evaluation.
Phase 5 – national profile	Common Icelandic OpenADR profile, certification, primacy rules and multi-DSO interoperability.

14.3 Meeting objective with Landsnet

The immediate meeting is best treated as institutional reconnaissance rather than technical design. The central question is whether Landsnet regards the four-peak static charge as a fundamental economic necessity or as a proxy that can evolve once DSO demand becomes measurably controllable. A positive answer does not require Landsnet to operate the RARIK pilot; it establishes that the upstream constraint need not remain static forever.

- Would Landsnet recognise verified delivery-point peak reduction from DSO-controlled flexibility as economically different from uncontrolled firm demand?
- Would Landsnet provide a constraint or peak-risk signal that RARIK could translate into local envelopes?
- Could a future tariff distinguish dynamic, curtailable DSO capacity from conventional capacity, building on existing curtailable-transmission principles?
- What telemetry, audit and performance evidence would Landsnet require before relying on an aggregated resource?
- Does Landsnet have the institutional mandate to pilot such a change, and which other stakeholder must approve it?

15. Counterarguments and Limitations

Counterargument	Assessment
“The grid must simply be built.”	True for persistent energy, thermal and reliability constraints. Incomplete because physical build times exceed charging deployment times and rare peaks can be managed more cheaply.
“Dynamic control transfers reliability risk to customers.”	It does if the product is unclear. A transparent firm/non-firm contract, service-level disclosure and safe fallback make the trade explicit.
“Forecasts are still required.”	Correct. Forecasts remain inputs for planning and allocation, but they should not be treated as immutable truth or sole basis for an upfront penalty.
“ICT and cyber costs may offset savings.”	They are real and must be included. International DOE work confirms enabling ICT and agent costs. Scale and shared platforms are therefore important [15, 16].
“Customers may not accept curtailment.”	Then they can purchase firm access and pay the associated cost. The proposal adds choice; it does not remove firm service.
“Dynamic access may be unfair.”	Allocation rules must be transparent: first-come, pro-rata, priority service, auction or regulated essential-service rules. Fairness is a design variable, not a reason to preserve static scarcity.
“OpenADR is not a real-time grid-control protocol.”	Correct for protection and sub-second control. It is appropriate for supervisory envelopes, events and seconds-to-minutes flexibility when local enforcement remains authoritative.
“The exact benefit is unproven.”	The 1 September test proves control reliability, not avoided capex. Benefit quantification follows through measured peaks, denied

Counterargument	Assessment
	requests and network studies.

15.1 Limits of the “unavoidable path” claim

The phrase “a dynamically controlled grid is unavoidable to avoid disaster” is rhetorically strong. Academically, the defensible form is: dynamic control is a necessary component of a feasible transition because the alternatives—connection rationing, static overbuilding, punitive peak pricing and delayed electrification—cannot by themselves deliver the required speed and efficiency. Dynamic control is necessary but not sufficient; generation, reinforcement, maintenance and regulation remain necessary.

16. Conclusions and Recommendations

Iceland’s present tariff and connection architecture was built for relatively predictable, one-directional and slowly changing loads. Its key variables—contracted power, representative daily utilisation, annual peak, producer size and consumer/producer category—are treated as static commitments. Electrified transport and distributed energy resources are different: their net power is intermittent, bidirectional and controllable.

The current system already contains evidence that flexibility has economic value. Landsnet uses a four-peak tariff, offers curtailable transmission and procures system services. RARIK credits smaller generation for reducing upstream withdrawal. Netmáli values “other benefit” in principle. What is missing is a standard, reliable and auditable operational interface that turns controllability into a recognised network product.

OpenADR 3.0 can provide that interface. It can register resources, report requests and availability, issue time-bounded events, subscribe to updates and return measured performance. It cannot replace protection, safety certification, connection controllers or financial market systems. The proposed architecture keeps those responsibilities separate.

The exact cost recovered by RARIK’s innmötunargjald cannot be determined from public tariff material because no granular allocation is disclosed. The fee is based on agreed export capacity, while connection is charged separately, and its historical design explicitly followed Landsnet. Dynamic dispatch therefore creates a legitimate cost-causation challenge: a resource with no firm export right and zero-export fallback should not automatically be priced as though the DSO has reserved its nameplate capacity continuously.

16.1 Priority recommendations

7. Define the September success criterion as closed-loop boundary control with fail-safe behaviour and reconciled audit logs.
8. Adopt an Icelandic OpenADR 3.0 application profile for capacity request, operating envelope, flexibility offer, activation, net-power report and exception.
9. Create separate contracts and tariffs for firm access and dynamic non-firm access.
10. Require RARIK to publish or submit to the regulator a cost-causation study for innmötunargjald.
11. Credit verified avoided Landsnet peak cost and local reinforcement deferral, while retaining charges for real connection, metering, cyber and shared-network services.
12. Pilot seasonal public charging because it makes the static-model defect measurable and societally relevant.
13. Add BESS/V2G and industrial resources after the interface proof to demonstrate two-sided flexibility.
14. Establish DSO–TSO primacy and anti-double-commitment rules before commercial aggregation.
15. Use persistent denied-capacity hours and measured congestion—not theoretical nameplate alone—as a primary reinforcement signal.
16. Position dynamic control as the fast operational bridge while physical generation and network investment proceed.

Closing statement

Iceland cannot finance, permit and physically build every possible electrification peak before the demand arrives. The viable alternative is not to stop building the grid; it is to operate existing and new assets dynamically, allocate capacity when and where it exists, and build reinforcement where measured scarcity proves it is continuously required.

References

[1] Landsnet Raforkuspá Landsnets 2024–2050 (electricity demand and supply outlook). https://www.landsnet.is/utgafa-og-samskipti/ritogskyrslur/raforkuspa-landsnets-2024-2050/ Accessed 4 Aug 2026.

[2] Landsnet Financial Statements 2025. Investment, in-feed refund and revenue-cap discussion. https://www.landsnet.is/library?itemid=d5661d2a-8911-4a76-87a0-26d693db05a2 Published Feb 2026.

[3] Landsnet Flutningsgjaldskrá: calculation of DSO capacity charge from four highest monthly 60-minute peaks. https://www.landsnet.is/gjaldskra/flutningsgjaldskra/ Accessed 4 Aug 2026.

[4] Landsnet Gjaldskrá og reiknivélar: tariff effective 1 April 2026. https://www.landsnet.is/gjaldskra/gjaldskra-og-reiknivelar/ Accessed 4 Aug 2026.

[5] RARIK Gjaldskrá fyrir innmötun raforku, version 46, effective 1 December 2025. https://www.rarik.is/gjaldskrar/gjaldskrar/gjaldskra-fyrir-innmotun-raforku Accessed 4 Aug 2026.

[6] RARIK Breyting á gjaldskrá fyrir innmötun raforku frá 1. apríl 2022. https://www.rarik.is/frettir/breyting-a-verdskra-fyrir-innmotun-raforku-fra-1-april-2022 5 Apr 2022.

[7] Government of Iceland Regulation No. 302/2022 amending Regulation No. 1040/2005; system contributions for new DSO connections. https://www.reglugerd.is/reglugerdir/eftir-raduneytum/idnadarraduneyti/nr/22998 24 Feb 2022.

[8] Icelandic DSOs Netmáli 1.0: terms for additional costs for new DSO connections and major investments. https://eu-assets.contentstack.com/v3/assets/blt5bf0023ca96cf071/blt0739f4d3df1b1bdd/6824993f3510b0699c844f16/Netm%C3%A1li_1.0_Ma%C3%AD2025_Undirrita%C3%B0.pdf Effective 15 May 2025.

[9] Veitur All electricity DSOs approve Netmáli 1.0. https://www.veitur.is/frettir/allar-dreifiveitur-rafmagns-samthykkja-netmala-1-0 2025.

[10] RARIK Raforkudreifing kostar meira: historical statement that the injection tariff largely followed Landsnet. https://www.rarik.is/frettir/raforkudreifing-kostar-meira-1 29 Jul 2009.

[11] Landsnet Niðurstaða Hæstaréttar skapar óvissu: Supreme Court result on transmission in-feed charge. https://www.landsnet.is/utgafa-og-samskipti/frettir/frett/?documentid=78a33a09-62b7-4ce9-b43d-ab977e31207a 7 Jun 2024.

[12] OpenADR Alliance Deep Dive in OpenADR 3.0: HTTP/JSON/OAuth/REST, programs, events, reports, resources and certification. https://www.openadr.org/assets/1_2_1_Deep%20Dive%20in%20OpenADR%203.0.pdf 2024/2025 material.

[13] OpenADR Alliance What you need to know about OpenADR 3.1.0. https://www.openadr.org/index.php?day=17&id=100%3Awhat-you-need-to-know-about-the-latest-version-of-openadr-3-openadr-3-1-0-&month=09&option=com_dailyplanetblog&view=entry&year=2025 17 Sep 2025.

[14] International Energy Agency Electricity 2026: Grids; comparison of 5–15 year grid lead times and 1–2 year EV charging deployment. https://www.iea.org/reports/electricity-2026/grids 2026.

[15] Australian Renewable Energy Agency Dynamic Operating Envelopes Workstream. https://arena.gov.au/knowledge-innovation/distributed-energy-integration-program/dynamic-operating-envelopes-workstream/ Accessed 4 Aug 2026.

[16] Australian Energy Regulator Export Limit Guidance Note: flexible export limits as an alternative to static limits. https://www.aer.gov.au/system/files/2024-10/Export%20Limits%20Guidance%20Note.pdf 2024.

[17] Energy Networks Association (UK) Good Practice Guide: Active Network Management. https://www.energynetworks.org/assets/images/Resource%20library/ANM%20Good%20Practice%20Guide%202015.pdf 2015.

[18] International Energy Agency Global EV Outlook 2024: trends in EV charging and MCS maximum power assumptions. https://www.iea.org/reports/global-ev-outlook-2024/trends-in-electric-vehicle-charging 2024.

[19] Icelandic Tourist Board (Ferðamálastofa) Foreign passenger departures through Keflavík, monthly data workbook and methodology. https://www.ferdamalastofa.is/is/gogn/fjoldi-ferdamanna/talningar-ferdamalastofu-i-flugstod-leifs-eiriksonar Workbook downloaded 4 Aug 2026.

[20] Icelandic National Energy Regulatory Authority Raforkueftirlitið: revenue limits based on permitted capital cost, operating cost and depreciation. https://uos.is/raforkueftirlitid Accessed 4 Aug 2026.

[21] Alþingi Committee report: transmission cost paid by DSOs is added to DSO revenue limits as actual cost. https://www.althingi.is/altext/pdf/155/s/0421.pdf Parliamentary report.

[22] RARIK Nýjar gjaldskrár taka gildi: front-loaded electrification investment and cost pressures. https://www.rarik.is/frettir/frettir/nyjar-gjaldskrar-taka-gildi 30 Apr 2025.

[23] RARIK Við einföldum ferli við tengingu minni virkjana: prior case-by-case profitability assessment and reservation of capacity. https://www.rarik.is/frettir/frettir/vid-einfoldum-ferli-vid-tengingu-minni-virkjana 1 Dec 2025.

[24] Government of Iceland Impact and Allocation Report 2024: transport as the largest contributor to Icelandic societal GHG emissions. https://www.stjornarradid.is/library/02-Rit--skyrslur-og-skrar/Impact%20and%20allocation%20report%202024%20vefur.pdf Published 2025.

Appendix A — Example OpenADR application payload semantics

The following is a semantic example rather than normative OpenADR JSON. Field names must be mapped to valid OpenADR 3.0 payload descriptors and agreed extensions during implementation.

```
SITE → RARIK VTN
resource: site-connection-001
currentNetPower_kW: 12
preferredImport_kW: 350
minimumUsefulImport_kW: 80
requiredEnergy_kWh: 120
latestCompletion: 2026-09-01T14:45:00Z
```

```
availableDownFlex_kW: 270

RARIK VTN → SITE
program: IS-DYNAMIC-CAPACITY-PILOT
event: envelope-00042
importMax_kW: 175
exportMax_kW: 0
validFrom: 2026-09-01T14:00:00Z
validUntil: 2026-09-01T14:10:00Z
fallbackImportMax_kW: 50
priority: mandatory-network-limit
```

Appendix B — Data calculations

Metric	Calculation	Result
20 kW RARIK injection fee	105,930 + 0.02 × 837,638	ISK 122,683/year excl. VAT
100 × 20 kW resources	100 × 122,682.76	ISK 12.27 million/year excl. VAT
Landsnet four-peak relief	1 MW × 9,786,291	ISK 9.786 million/year
Single selected peak relief	(1 MW / 4) × 9,786,291	ISK 2.447 million/year
Tourism monthly peak/min	312,964 / 120,016	2.61×
Summer/winter average	282,941 / 130,088	2.18×

Appendix C — Glossary

Term	Definition
ANM	Active Network Management.
BESS	Battery Energy Storage System.
CPO	Charge Point Operator.
DER	Distributed Energy Resource.
DOE	Dynamic Operating Envelope.
DSO	Distribution System Operator.
EMS	Energy Management System.
EVSE	Electric Vehicle Supply Equipment.
MCS	Megawatt Charging System.
OCPP	Open Charge Point Protocol.
OpenADR	Open Automated Demand Response.
TSO	Transmission System Operator.
VEN	Virtual End Node in OpenADR.
VTN	Virtual Top Node in OpenADR.
V2G	Vehicle-to-Grid.
Innmötunargjald	Distribution-network charge applied to electricity injection.
Kerfisframlag	System contribution / additional connection contribution.